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Price spread prediction in high-frequency pairs trading using deep learning architectures

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ABSTRACT

High-Frequency Trading (HFT) leverages advanced algorithms and high-speed data transmission to execute a large volume of trades within extremely short timeframes and generate profit. However, predicting stock prices in this context is challenging due to the frequent fluctuations in bid and ask conditions within financial markets. Institutional investors often employ pairs trading strategies to mitigate the systemic risk associated with single products, simultaneously buying and selling highly correlated financial instruments to profit from fluctuations in abnormal price spreads. In this study, we utilized intraday continuous trading data from the Taiwan Stock Exchange, integrating commonly used stock market features relevant to HFT. By employing XGBoost for feature selection and combining it with deep learning models, we aimed to predict the relationship between price spreads and boundaries in pairs trading, thereby generating entry and exit signals. Although accurately predicting the relationship between price spreads and boundaries presents significant challenges, the model effectively learned the pattern of price spread changes. Applying the model's entry and exit signals in pairs trading demonstrated that this strategy can enhance win rates and achieve stable profits in the volatile intraday market environment. This research provides practical implications for those interested in high-frequency trading and deep learning models in the financial market, equipping them with valuable knowledge and insights.

1. Introduction

High-Frequency Trading (HFT) is a trading strategy that leverages advanced computer algorithms and high-speed data transmission technologies to capitalize on price spreads through a large volume of frequent trades in volatile markets (Dutta, Karpman, Basu, and Ravishanker, 2023; Petukhina, Reule, and Härdle, 2021). Due to the substantial costs associated with the necessary equipment and investments, HFT is predominantly employed by institutional investors. The key characteristics of HFT include rapid execution, large trade volumes, short-term investments, and high liquidity of the traded assets. This strategy enables institutional investors to avoid holding positions for extended periods, thereby minimizing potential losses while accumulating profits with relatively low risk. HFT is widely utilized across various financial markets, including the stock market (Van Kervel and Menkveld, 2019), futures market (Liang, Xu, Hu, and Du, 2023), foreign exchange market (Rostamian and O'Hara, 2022), and cryptocurrency market (Chen, Li, and Sun, 2020). Numerous studies have introduced algorithms for HFT data, such as Deep Learning (Kumar and Haider, 2021; Tashiro, Matsushima, Izumi, and Sakaji, 2019) and Reinforcement Learning (Briola, Turiel, Marcaccioli, and Aste, 2021; Lee, Koh, and Choe, 2021), to predict asset price trends and generate trading decisions. These algorithms have demonstrated exemplary performance in backtesting across multiple financial market products, instilling confidence in the robustness of our research methodology and providing reassurance to the financial market community.

Institutional investors utilize pairs trading strategies for hedging to mitigate the systemic risk of investing in a single product (Keshavarz Haddad and Talebi, 2023). Pairs trading is a market-neutral strategy that remains insulated from overall market movements. Traders simultaneously buy and sell two or more highly correlated financial products, such as stocks, futures, or other derivatives. The primary source of profit in pairs trading arises from the temporary divergence or convergence of abnormal price spreads (Brim, 2020; Kim, Park, and Lee, 2022), making spread prediction the foremost goal of this strategy.

Recently, scholars have integrated pairs trading with high-frequency rebalancing using Matlab to analyze cryptocurrency exchange markets across various trends and volatility regimes (Bağcı and Soylu, 2024).

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Additionally, the authors explore pairs trading in cryptocurrencies using statistical approaches at 1-min and 5-min frequencies (Ko et al., 2024). However, while these studies address high-frequency pairs trading, they often overlook the importance of evaluating each feature before optimal portfolio selection, relying on traditional methods for their analyses. Xu and Luo (2023) have proposed applying reinforcement learning methods to select appropriate trading pairs in financial markets. However, their focus is primarily on pair selection rather than feature selection, with limited emphasis on high-frequency trading.

To examine the significance and robust profitability of pairs trading, this study adopts an institutional investor's perspective by utilizing the intraday continuous trading dataset provided by the Taiwan Stock Exchange. This dataset includes limit order book (LOB) records for all instruments traded on the Taiwan stock market. Unlike the daily trading data consolidated by major stock market information websites—where each instrument is typically represented by only one record per trading day, potentially overlooking significant intraday trading nuances—the LOB records encompass all buy and sell orders and transactions occurring during intraday trading. Consequently, each instrument may generate millions of records per trading day, enabling a more granular and comprehensive analysis and facilitating the extraction of implicit information derived from the intraday order behavior of market participants.

The contributions of this study are as follows: First, we integrate intraday continuous trading data from the Taiwan stock market to extract three types of market features: basic features, LOB features, and technical indicators. A feature selection module is employed to filter out unnecessary noise, retaining only the essential features that significantly contribute to predicting the price of the target instrument. To apply pairs trading to instruments in the Taiwan stock market, the pair selection module utilizes correlation and cointegration tests to identify instrument pairs with strong correlations and cointegration relationships suitable for pairs trading. Finally, the price spread prediction module leverages various deep learning models to predict the relationship between price spreads and boundaries of the selected instruments, using the most accurate predictions as signals for entry and exit in pairs trading. These signals provide institutional investors with a reliable reference for executing pairs trading, enabling stable profits in the volatile intraday trading environment.

2. Related works

2.1. *High-frequency trading*

High-frequency trading (HFT) involves executing many trades rapidly to profit from small price spreads. HFT participants rely on advanced technologies and equipment to analyze trading data and execute trades at extremely high speeds (Van Kervel and Menkveld, 2019). Given the necessity of conducting numerous transactions, HFT typically occurs in markets with high trading activity, such as the stock, futures, forex, and cryptocurrency markets. While stock markets are constrained by trading hours, forex and cryptocurrency markets operate 24 h daily, resulting in more stable prices and market characteristics. In this study, we draw on literature from the stock and futures markets and examine studies from the forex and cryptocurrency markets to observe the differences in HFT strategies and frameworks across these various markets.

In the context of the stock and futures markets, Malceniece, Malcenieks, and Putniņš (2019) utilized data from Chi-X across 12 European stock markets and found that HFT increased the liquidity and comovement of small and mid-cap stocks in Europe. Zhang, Huang, Xu, and Xing (2023a) proposed a strategy to reduce computational costs through feature extraction, demonstrating that support vector machines and elastic nets (ENet) are well-suited for HFT. Liang et al. (2023) successfully predicted high-frequency data in the Chinese futures market using wavelet transforms and Daubechies wavelets in deep learning models, thereby constructing high-return, risk-controlled trading strategies. To further enhance HFT speed, Li, Shen, and Qian (2023) introduced the Online LGT (O-LGT) framework, showcasing its significant computational speed advantage over other neural network models using CSI 500 futures index data. Ye, Yang, and Chen (2024) innovatively converted LOB data into images, demonstrating that image-based models using convolutional neural network (CNN) achieved high F1 scores across different image sizes and outperformed traditional machine learning models, proving superior predictive capabilities for trading strategies in backtesting. Korniejczuk and Ślepaczuk (2024) proposed a statistical arbitrage strategy based on graph clustering algorithms, combining quantitative methods with machine learning classifier ensembles to enhance risk-adjusted returns and mitigate the impact of transaction costs on profits. Their experimental results indicated that this strategy outperformed relevant benchmark methods when considering transaction costs and excelled in trade signal detection and risk management.

In the forex market, Rostamian and O'Hara (2022) introduced the Directional Change (DC) method, which combined CNN and long short-term memory (LSTM) models to enhance predictive performance for currency pair price movements. Rundo (2019) developed a high-frequency trading system using LSTM and reinforcement learning, achieving an impressive investment return rate of 98.23%. Zafeiriou and Kalles (2023) created an ultra-high-frequency trading system that utilizes neural networks for real-time market decision-making, demonstrating profitability in real-data testing.

In the cryptocurrency market, Fang et al. (2021) combined autoencoders with a progressive retraining approach to enhance the predictive capability of LSTM models for high-frequency trading data. Ozer and Sakar (2022) proposed a trading system that improved investment returns through outlier detection, demonstrating that increasing trading frequency could enhance profitability. He, Zheng, and Yang (2023) applied deep reinforcement learning (DRL), integrating tick data and predictive signals, which improved strategy performance in cryptocurrency high-frequency trading while effectively managing risk. Zong et al. (2024) introduced a high-frequency trading method based on a Markov Decision Process (MDP) structure, successfully addressing market volatility to achieve maximum profitability and excellent risk management. Qin et al. (2024) developed a three-stage hierarchical reinforcement learning framework specifically designed for high-frequency trading, significantly outperforming many robust baseline methods in cryptocurrency market experiments. Bağcı and Soylu (2024) proposed the High-Frequency Rebalancing Architecture (HFRA), which significantly increased profit potential during high-volatility uptrends and reduced potential losses during downtrends. Compared to periodic rebalancing and threshold rebalancing strategies, HFRA demonstrated higher profitability in long-term applications.

These studies showcase the potential and challenges of high-frequency trading in various financial markets, exploring the application of multiple models and methods in high-frequency data prediction and trading strategies.

2.2. *Pairs trading*

Pairs Trading is a quantitative strategy designed to profit from the price spreads between two or more financial instruments. This strategy involves simultaneously trading two sets of instruments—such as stocks, indices, forex, and other financial products—that exhibit a certain level of correlation. By shorting one set and going long on the other, pairs trading aims to capitalize on abnormal price spreads resulting from temporary market fluctuations (Brim, 2020; da Matta, 2020). As a market-neutral investment strategy, pairs trading focuses on risk management rather than maximizing profits. In highly volatile high-frequency trading, pairs trading enables investors to adjust asset allocation according to risk tolerance levels, achieving optimal investment outcomes (Lu et al., 2022; Sarmento and Horta, 2020).

Fuster and Zou (n.d.) conducted cointegration tests using data from the top 50 S&P 500 constituent stocks, finding that predicting spread trends is more feasible than predicting actual values, with neural networks outperforming traditional models in spread prediction. Ospino, Fernandez, and Carchano (2020) combined LSTM and MLP models to predict trend signals for S&P 500 constituent stocks, demonstrating that neural networks can enhance the robustness and profitability of pairs trading strategies. Flori and Regoli (2021) utilized LSTM models to predict relative market returns among peer stocks and improved pairs trading performance. Su, Lai, Shih, Wang, and Huang (2022) employed data from the Taiwan Stock Exchange to prove their deep learning architecture could accurately predict spread trends. Kuo, Chang, Dai, Chen, and Chang (2022) proposed a two-stage deep learning model to improve the investment performance of pairs trading strategies. Their model, trained on a time-invariant spread process, could more consistently enhance investment performance by optimizing trigger threshold selection and eliminating unprofitable stock pairs.

Furthermore, numerous studies have confirmed that pairs trading can yield profits with low risk. Cerda, Rojas-Morales, Minutolo, and Kristjanpoller (2022) employed the ACO-PT algorithm to optimize pairs trading dynamically, maintaining high daily returns and Sharpe Ratios. Brim (2020) utilized a Double Deep Q-network (DDQN) to learn pairs trading strategies for stocks, incorporating a negative reward multiplier to encourage the DDQN agent to take more conservative actions, thereby increasing cumulative returns. Baek, Glambosky, Oh, and Lee (2020) applied a Support Vector Machine (SVM) for futures pairs trading, achieving high returns in high-risk markets. Chang, Man, Xu, and Hsu (2021) used LSTM models to predict adjusted closing prices of qualifying instruments, dividing the portfolio into aggressive and defensive investment portfolios, with aggressive stocks showing superior performance. Wang, Sandås, and Beling (2021) defined a reward function to enable reinforcement learning models to achieve higher returns while controlling risk. Lu et al. (2022) proposed the SAPT strategy to address market structural breaks, which performed exceptionally well during COVID-19. Ti, Dai, Wang, Chang, and Sun (2024) introduced a pairs trading strategy based on deriving asymptotic means to determine opening and closing thresholds, demonstrating significant performance improvement and reduced trading risk in backtests. Vergara and Kristjanpoller (2024) developed a trading system based on deep reinforcement learning (DRL) to generate arbitrage portfolios, extract market signals, and make trading decisions. Their results indicated that this strategy remained profitable after accounting for trading costs, outperforming basic arbitrage strategies and other models regarding total returns, risk, and risk-adjusted returns.

These studies demonstrate that pairs trading strategies can yield profits while effectively managing risk, enabling investors to achieve returns even in volatile markets. However, the ability to accurately identify the timing of abnormal price spreads, execute timely entry and exit points, and balance profits with risk remains crucial for investors.

3. The proposed framework

This study adopts the perspective of institutional investors, focusing on achieving stable returns rather than maximizing profits. In the context of high-frequency continuous trading, pairs trading is recognized as a robust strategy. Our framework is structured into three key components, as illustrated in Fig. 1. The first component is the feature selection module, which analyzes the intraday market features of the instruments. Utilizing the XGBoost module, we assess the importance of each feature in predicting stock prices to select those that best represent the instruments. The second component is the pair selection module, which applies statistical tests to filter and identify eligible pairs of instruments. The third component is the price spread prediction module, which forecasts the trend of price spread changes based on the pairs identified by the pair selection module, monitoring the divergence and convergence of spreads and using the results as entry and exit signals. Finally, a pairs trading backtest is conducted to evaluate whether this framework can provide traders with effective entry and exit signals and generate profits.

3.1. Feature selection module

We utilize intraday continuous trading data and classify market features into three types: basic features, limit order book features, and technical indicators. To mitigate the potential issue of overly homogeneous features affecting model predictions, XGBoost is implemented as the feature selection module. A three-class classification task is designed to predict the mid-price movement—upward, downward, or unchanged. By evaluating the importance of each feature through XGBoost, we select those that significantly impact mid-price movement prediction. These selected features are then used as input criteria for predicting pair price spreads.

3.1.1. Basic features

We utilize intraday trading records to consolidate basic features for each time interval, including the opening price (Open), highest price (High), lowest price (Low), closing price (Close), and trading volume (Volume). These serve as the fundamental attributes for each time segment of the instrument.

3.1.2. Limit order book features

We acknowledge that certain features, particularly trade volume imbalance, best-order imbalance, queue imbalance, and others, have proven effective in high-frequency trading within cryptocurrency markets (Schnaubelt, 2022). However, the Taiwan stock market operates within a stable and regulated framework,[1] with fixed trading hours and a 10% daily price fluctuation limit, resulting in lower volatility compared to 24/7 cryptocurrency markets without price caps. Taiwan's regulatory framework closely monitors trading behaviors,[2] flagging large-volume trades and significant price changes to prevent manipulation. In contrast, cryptocurrency markets, with looser regulatory oversight, rely on these features to detect rapid price shifts and effectively manage manipulation risks.

Limit order book features are derived from buyers' and sellers' order and transaction records during intraday trading, as supported by recent literature (Barez, Bilokon, Gervais, and Lisitsyn, 2023; Bilokon and Qiu, 2023; Fang et al., 2024; Kolm, Turiel, and Westray, 2023; Schnaubelt, 2022; Zhang, Huang, Xu, and Xing, 2023b). A summary of these features is provided in Table 1.

- Best bid (ask) price

In the order records for all buyers (sellers), the highest price that buyers are willing to pay is designated as the best bid price, while the lowest price that sellers are willing to accept is defined as the best ask price.

- Average best bid (ask) price

The average *Best Bid (Ask) Price*, calculated over n time points, reflects the changes in the *Best Bid (Ask) Price* over a specific period. The average *Best Bid (Ask) Price* is computed as follows:

$$\text{Avg Best Bid(Ask) Price}_n = \frac{1}{n}\sum_{t=1}^{n}\text{Best Bid(Ask) Price}_t \tag{1}$$

where $\text{Avg Best Bid(Ask) Price}_n$ represents the average Best Bid (Ask)

[1] https://www.twse.com.tw/en/about/company/guide.html

[2] https://twse-regulation.twse.com.tw/m/en/LawContent.aspx?FID=FL007225

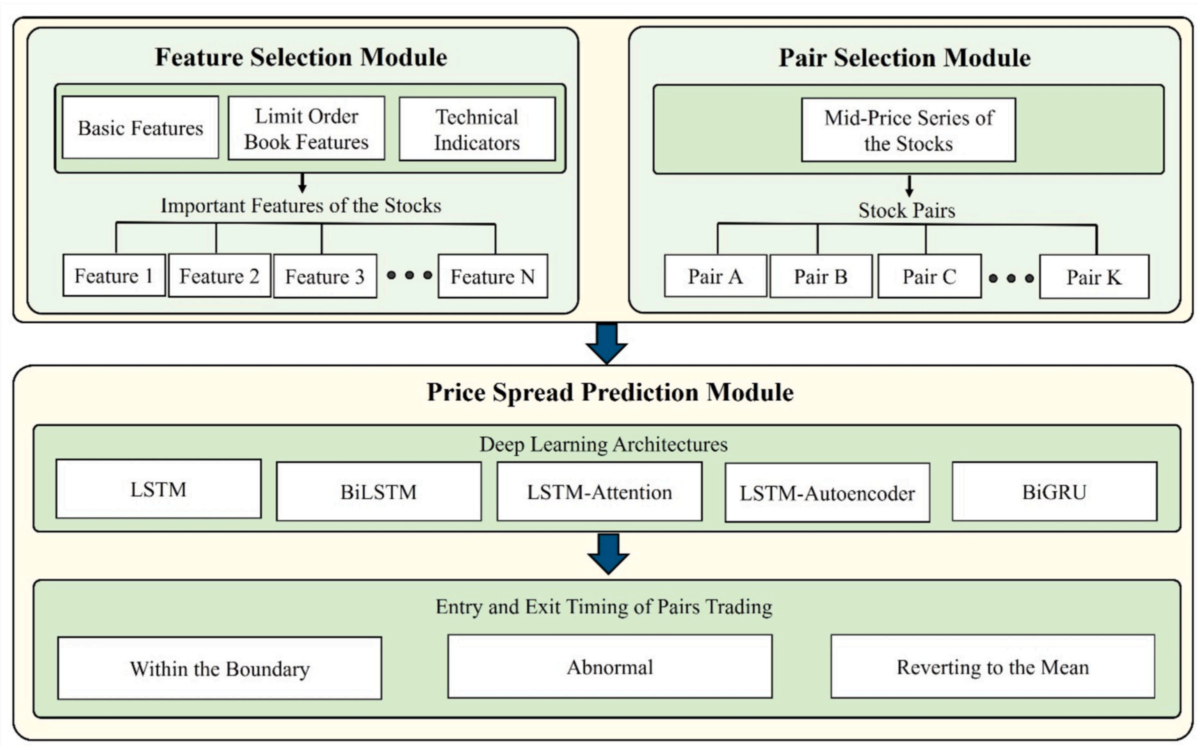

Fig. 1. Framework.

Table 1
Limit order book features.

Feature	Description	References
Best bid/Ask price	Best Bid Price: the highest price buyers are willing to pay. Best Ask Price: the lowest price sellers will accept.	(Fang et al., 2024; Kolm et al., 2023; Zhang et al., 2023b)
Average best bid/Ask price	The average best bid or ask price over a defined period.	(Zhang et al., 2023b)
Mid-price	The equilibrium price in the market, calculated as the midpoint between the best bid and ask prices.	(Barez et al., 2023; Bilokon and Qiu, 2023; Fang et al., 2024; Schnaubelt, 2022; Zhang et al., 2023b)
Mid-price change	The variation in mid-price over each time interval, tracking short-term price movement.	(Barez et al., 2023; Bilokon and Qiu, 2023; Fang et al., 2024; Schnaubelt, 2022; Zhang et al., 2023b)
Bid-ask spread	The relative difference between the best bid and ask prices, an indicator of market liquidity.	(Fang et al., 2024; Kolm et al., 2023; Schnaubelt, 2022)

price over n time points, and $\text{Best Bid(Ask) Price}_t$ denotes the Best Bid (Ask) price at the t-th time point.

- Mid-price

The mid-price is calculated using the best bid and ask prices for each time interval. It represents the equilibrium price in the market, providing a more accurate reflection of current market conditions. The calculation is as follows:

$$\text{Mid Price}_t = \frac{P_{bid,t} + P_{ask,t}}{2} \tag{2}$$

where $P_{bid,t}$represents the best bid price at time point t, and $P_{ask,t}$ represents the best ask price at time point t.

- Mid-price change

The mid-price change for each time interval is calculated as follows:

$$\text{Mid Price Change}_t = \text{Mid Price}_t - \text{Mid Price}_{t-1} \tag{3}$$

where Mid Price_t represents the mid-price at time point t and Mid Price_{t-1}represents the mid-price at the previous time point t-1. By subtracting the mid-price of the previous time point from the current mid-price, we can determine the change in the mid-price compared to the previous interval.

- Bid-ask spread

He et al. (2023) noted that the bid-ask spread serves as an indicator of market liquidity. A narrower spread suggests that the bid and ask prices are closer together, increasing the likelihood of a transaction and signaling higher market liquidity. Conversely, a wider spread indicates lower market liquidity. The calculation is as follows:

$$BAS_t = \frac{(P_{ask,t} - P_{bid,t})}{P_t^{bid}} \tag{4}$$

where BAS_t represents the bid-ask spread at time point t, $P_{ask,t}$ represents the best ask price at time point t, and $P_{bid,t}$ represents the best bid price at time point t.

- Bid (ask) volume

Bid (ask) order volume represents the total number of shares bought (sold) by all buyers (sellers) placing orders at a specific time point. The formula is as follows:

$$\text{Volume}_{Bid(Ask),t} = \sum_{p=1}^{n} \text{Vol}_{Bid(Ask),p} \tag{5}$$

where n represents the number of buyers (sellers) at time point t, and

$Vol_{Bid(Ask),p}$ denotes the order volume of the p-th buyer (seller) at time point t.

- Bid (ask) depth, trading depth

Trading depth represents the total number of quotes from buyers and sellers at a specific time. It is defined as follows:

$$\text{Trading Depth}_t = D_t^{bid} + D_t^{ask} \tag{6}$$

where D_t^{bid} represents the number of bids (bid depth) at time point t, and D_t^{ask}represents the number of asks (ask depth) at time point t. Trading depth offers insights into the instrument's liquidity at a given time and aids in understanding short-term price movements.

- Trading volatility

Trading volatility measures the standard deviation of the mid-price over a specified period, indicating the current level of market volatility and associated implied risk. The formula is as follows:

$$\text{Trading Volatility}_n = \sqrt{\text{Var}(X_n)} \tag{7}$$

$$\text{Var}(X_n) = \frac{1}{n}\sum_{t=1}^{n}(p_t - \mu_n)^2 \tag{8}$$

where $\text{Trading Volatility}_n$ represents the trading volatility over n time points, $\text{Var}(X_n)$ denotes the variance over n time points, P_t represents the mid-price at time point t, and μ_n is the average mid-price over the n time points.

3.1.3. Technical indicators

- Exponential moving average (EMA)

EMA incorporates the concept of weighting, placing greater importance on the most recent prices, thereby increasing their influence on the average. This makes the EMA more responsive to market fluctuations. The formula for calculating the EMA is as follows:

$$EMA_t = \alpha \times \text{Close}_t + (1-\alpha) \times EMA_{t-1} \tag{9}$$

where $EMA_t$is the EMA at time point t, α is the smoothing factor, typically defined as $\frac{2}{N+1}$, where N is the period for calculating the EMA, and Close_t represents the closing price at time point t.

- Relative strength index (RSI)

RSI is a technical indicator used to evaluate the strength of buying and selling pressures. When buying pressure is weak, prices tend to decline, whereas prices tend to rise when selling pressure weakens. The calculation formula for RSI consists of two parts:

$$RS = \frac{\text{Average Gain}_N}{\text{Average Loss}_N} \tag{10}$$

$$RSI = 100 - \frac{100}{1+RS} \tag{11}$$

where Average Gain_N(Average Loss_N) refers to the average price increases (decreases) over the specified N periods. When the *RSI* exceeds 70, it indicates an overbought condition, suggesting that the market may be overheating and signaling a potential selling opportunity. Conversely, when the *RSI* falls below 30, it indicates an oversold condition, suggesting that the market may be cooling down and signaling a potential buying opportunity.

- Stochastic oscillator (KD)

The KD indicator is a technical analysis tool used to identify overbought and oversold conditions in the market. It consists of two lines: the K value, which represents the fast stochastic indicator (also known as the fast line), and the D value, which represents the slow stochastic average (also known as the slow line). The KD indicator is calculated using the Raw Stochastic Value (RSV), which indicates whether the current stock price is strong or weak relative to stock prices over the past N periods. The formula for calculating RSV is as follows:

$$RSV_t = \frac{\text{Closing Price}_t - \text{Lowest Price}_N}{\text{Highest Price}_N - \text{Lowest Price}_N} \times 100 \tag{12}$$

where Closing Price_t represents the closing price at time t, and Lowest Price_N and Highest Price_N represent the lowest and highest prices over the last N periods, respectively. The formulas for calculating the K value and D value are as follows:

$$K_t = K_{t-1} \times \frac{2}{3} + RSV_t \times \frac{1}{3} \tag{13}$$

$$D_t = D_{t-1} \times \frac{2}{3} + K_t \times \frac{1}{3} \tag{14}$$

The KD indicator is primarily evaluated based on the K value. If the K value exceeds 80, it signals overbought conditions, indicating that the stock price is strong and likely to rise. Conversely, if the K value falls below 20, it signals oversold conditions, suggesting that the stock price is weak and likely to decline.

- Bollinger bands

Bollinger Bands are an extension of the Moving Average (MA) concept, primarily used to measure price volatility and identify potential high and low price levels in the market. Bollinger Bands consist of three components: the upper band, the middle band, and the lower band. The calculation formulas are as follows:

$$\text{Middle Band} = MA_n \tag{15}$$

$$\text{Upper (Lower) Band} = \text{Middle Band} \pm K \times \sigma_n \tag{16}$$

where MA_n is the simple moving average of the closing prices over n time points, σ_n represents the standard deviation of the closing prices over n time points, and K determines the width of the Bollinger Bands. In this study, K is set to 1.5. When the instrument's price touches the upper band, it suggests that the market may be overbought. Conversely, the market may be oversold when the price touches the lower band.

3.2. Pair selection module

In this study, the selection of pairs is guided by the Pearson correlation coefficient, which quantifies the relationship between the mid-prices of the instruments. Pairs exhibiting a correlation exceeding 0.9 are shortlisted for further statistical testing.

The Augmented Dickey-Fuller (ADF) test (Cheung and Lai, 1995) is initially employed to assess the stationarity of the mid-price time series. For non-stationary series, a cointegration test is subsequently performed. Specifically, the Engle-Granger two-step approach (Chang et al., 2021) is utilized to determine whether a cointegration relationship exists between the two non-stationary time series. Such a relationship suggests a long-term equilibrium between the series, rendering them suitable for pairs trading. Consequently, pairs displaying cointegration are selected for price spread prediction to extract insights from the intraday trading data.

3.3. Price spread prediction module

The price spread prediction module employs the critical features identified during the preceding feature selection process as inputs to forecast the price spread trend between the two instruments for the subsequent time interval. We adopt the pairs trading activation threshold specified by Chang et al. (2021). The entry condition for initiating a pairs trade is triggered when the price spread between the two instruments exceeds the mean price spread by ±1.5 times the standard deviation of the price spread. The exit condition occurs when the price spread reverts to the mean. The formulas for calculating the mean price spread, the standard deviation of the price spread, and the corresponding upper and lower bounds are presented as follows:

$$\mathrm{Mean}_t = \sum_{i=t-s}^{t} \mathrm{Price\ Spread}_i \tag{17}$$

$$\mathrm{Std}_t = \sqrt{\frac{1}{n}\sum_{i=t-s}^{t} (\mathrm{Price\ Spread}_i - \mathrm{Mean}_t)^2} \tag{18}$$

$$\mathrm{Upper(Lower)\ Bound}_t = \mathrm{Mean}_t \pm 1.5 \times \mathrm{Std}_t \tag{19}$$

where Mean_t represents the mean mid-price spread between the two instruments at time *t*. $\mathrm{Price\ Spread}_i$ denotes the price spread between the two instruments at time *i*, where *i* ranges from the starting time point *s* to *t*. Std_n denotes the standard deviation of the price spread between the two instruments at time *t*. $\mathrm{Upper\ Bound}_t$ and $\mathrm{Lower\ Bound}_t$ represent the upper and lower limits of the price spread at time *t*, respectively.

We formulate the model's prediction task as a three-class classification problem: (1) the price spread remains within the boundaries, (2) an abnormal price spread occurs (where the price spread touches or exceeds the upper or lower boundary), and (3) the price spread reverts to the mean. By leveraging deep learning models to forecast the price spread trend and considering the distance to the boundaries, we aim to identify whether the spread diverges unusually or converges, thus generating entry and exit signals for pairs trading.

In line with the concept of dynamic boundaries introduced by Lu et al. (2022), we use a sliding window to calculate the boundaries of the mid-price spread to reflect the most recent dynamics of the price spread. This approach allows for a more flexible adjustment of the boundaries based on market volatility, enhancing the opportunity for trading profits.

To facilitate the model's ability to learn the relationship between the current price spread and the boundaries, we introduce three additional features in the price spread trend prediction: the distance from the current price spread to the upper boundary, the distance to the lower boundary, and the distance to the mean price spread. The formulas for calculating these features are provided as follows:

$$\mathrm{Dist}_t^{\mathrm{Upper}} = \mathrm{Price\_Spread}_t - \mathrm{Upper\ Bound}_t \tag{20}$$

$$\mathrm{Dist}_t^{\mathrm{Lower}} = \mathrm{Price\_Spread}_t - \mathrm{Lower\ Bound}_t \tag{21}$$

$$\mathrm{Dist}_t^{\mathrm{Mean}} = \mathrm{Price\_Spread}_t - \mathrm{Mean}_t \tag{22}$$

where $\mathrm{Price\_Spread}_t$ represents the mid-price spread between two instruments at time t. The upper boundary, lower boundary, and mean price spread at time t are denoted as $\mathrm{Upper\ Bound}_t$、$\mathrm{Lower\ Bound}_t$、Mean_t, respectively.

We utilize five deep learning architectures to predict price spread trends: Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), LSTM with Attention Mechanism (LSTM-Attention), LSTM with Autoencoder (LSTM-Autoencoder), and Bidirectional Gated Recurrent Unit (BiGRU), which is widely applied in time series forecasting. Each model is trained 100 times per instrument pair to ensure robustness. The best-performing model is selected based on its prediction accuracy, and the results are analyzed to identify the most suitable architecture for forecasting the price spread trend. Predictions from the optimal model are then used to generate entry and exit signals for pairs trading.

4. Experiments

4.1. Dataset

We utilized intraday continuous trading data from the Taiwan Stock Exchange, covering the period from April 1, 2020, to March 31, 2021. This dataset includes all order and transaction records for the listed instruments. During the COVID-19 pandemic, the shipping & transportation, and semiconductor industrial groups emerged as popular trading targets (highlighted in yellow in Fig. 2). To mitigate slippage and address potential liquidity constraints during high-frequency trading, we selected the top 10 instruments by trading volume within these sectors. Minute-by-minute and five-minute interval data were used for further analysis (as presented in Table 2). The dataset is divided into two periods: the formation period (April 1, 2020, to February 23, 2021), which is used for model training, and the backtesting period (February 24, 2021, to March 31, 2021), which is employed for trading backtesting.

4.2. Data preprocessing

4.2.1. Time frequency selection and missing value handling

We calculated and recorded the ratio of missing values for each instrument's minute-by-minute and five-minute mid-price data (Table 3). It was noted that the minute-by-minute data exhibited a high percentage of missing values, suggesting that this time granularity was too fine, with numerous time points lacking corresponding order and transaction records. This could adversely affect the experimental results. To ensure data integrity, we used five-minute intervals for further analysis.

To minimize the impact of missing values on the experiment, instruments with a missing rate exceeding 4% in the five-minute data were excluded (highlighted in gray in Table 2). As a result, 18 instruments were selected for further analysis: UMC (2303), TSMC (2330), Macronix (2337), WEC (2344), SiS (2363), SUNPLUS (2401), NTC (2408), KYEC (2449), PANJIT (2481), EMC (2603), SNC (2605), U-MING (2606), YMTC (2609), CAL (2610), WANHAI (2615), EVAAIR (2618), THSRC (2633), and ASX (3711). These 18 securities, based on five-minute interval data, were utilized for the subsequent analysis.

4.2.2. Sliding window

According to Arian, Mobarekeh, and Seco (2024), a key advantage of walk-forward cross-validation (WFCV) in backtesting is its realistic simulation of live trading. By training and testing on sequential data segments, WFCV offers a robust measure of predictive power and adaptability in dynamic financial markets. However, its computational intensity, especially with high-frequency data and complex models, as well as the sensitivity to window size selection, presents challenges for certain datasets.

To address these limitations, we incorporated a sliding window technique during the model training phase to generate feature aspects. This method adapts effectively to real-time market dynamics, enhancing computational efficiency through rapid, low-latency updates. It enables the model to capture short-term stock market fluctuations and predict the relationship between the price spread and its upper and lower bounds at subsequent time points.

We employ a sliding window approach (Geng, Li, and Sun, 2023) to aggregate data into time segments, capturing stock market features to reflect actual market dynamics. This method enables the model to more effectively observe changes at each time point, enhancing the accuracy and flexibility of predictions. The sliding window technique defines two key parameters: window size and step size. Window size determines the

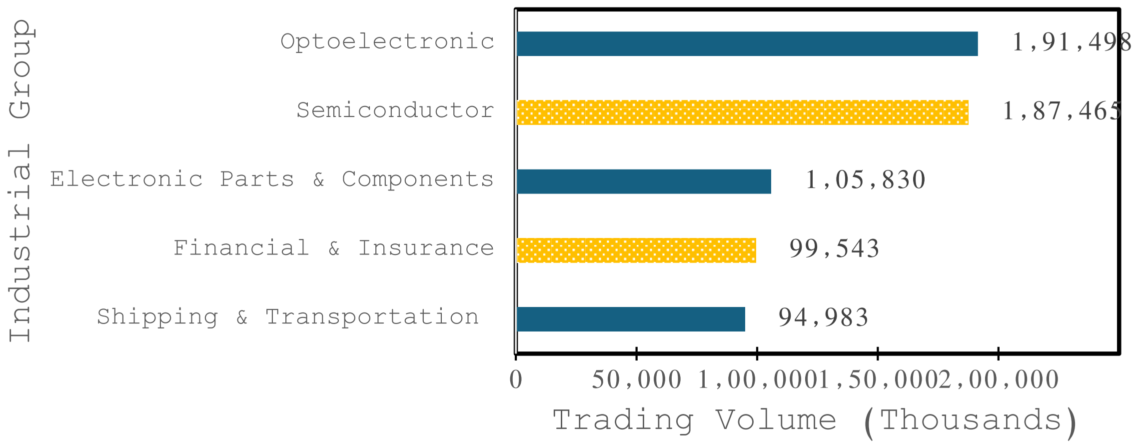

Fig. 2. Trading Volume by Industry Group (Top Five, April 1, 2020 – March 31, 2021).

Table 2
Top 10 stocks by trading volume in selected industrial groups.

Semiconductor			Shipping & transportation		
Code	Name	Trading volume (Thousands)	Code	Name	Trading volume (Thousands)
2303	UMC	46,315.738	2603	EMC	37,772.725
2344	WEC	21,644.421	2610	CAL	16,227.067
2337	Macronix	13,659.167	2609	YMTC	15,584.431
2330	TSMC	12,384.494	2618	EVAAIR	8461.026
3711	ASX	5077.342	2615	WANHAI	5100.330
2449	KYEC	4860.502	2605	SNC	2025.749
2408	NTC	4503.984	2633	THSRC	1716.965
2363	SiS	4470.996	2606	U-MING	1346.849
2481	PANJIT	3868.096	5608	S. W.	1089.380
2401	SUNPLUS	3711.231	2607	EITC	1080.642

Table 3
Missing data statistics.

Code	Count (per minute)	Missing rate (per minute)	Count (per 5 minutes)	Missing rate (per 5 minute)
2607	24,370	37.69%	741	5.73%
5608	23,124	35.76%	587	4.54%
2605	11,426	17.67%	85	0.66%
2609	9248	14.3%	87	0.67%
2615	7946	12.29%	63	0.49%
2606	7666	11.86%	19	0.15%
2401	6142	9.5%	31	0.24%
2633	1092	1.69%	0	0.0%
2481	996	1.54%	0	0.0%
2363	953	1.47%	3	0.02%
2603	202	0.31%	0	0.0%
2610	189	0.29%	0	0.0%
2618	162	0.25%	0	0.0%
3711	71	0.11%	0	0.0%
2449	50	0.08%	0	0.0%
2344	12	0.02%	0	0.0%
2408	7	0.01%	0	0.0%
2337	3	0.0%	0	0.0%
2303	0	0.0%	0	0.0%
2330	0	0.0%	0	0.0%

amount of data in each window, while step size specifies the number of time points the window advances after each iteration. In this study, we set the window size to 5 and the step size to 1, continuing this method across the entire dataset, encompassing both training and testing data. An illustration of the sliding window method is provided in Fig. 3.

4.2.3. Data normalization

We apply Min-Max normalization to scale all data values to a range between 0 and 1. However, this method can be sensitive to outliers, which may disproportionately compress the data if normalization is applied across all periods simultaneously, introducing the risk of look-ahead bias. This compression may also hinder the model's ability to accurately capture the real-time changes in stock market features.

To address this, we employ a sliding window technique, ensuring that Min-Max normalization is applied independently within each window. This ensures that only the data observed up to a given point in time t is used for normalization, thereby eliminating any future data influence and preventing lookahead bias. By restricting the normalization to each sliding window, the model is better equipped to capture feature changes that are reflective of actual market dynamics.

Moreover, to enhance the model's ability to integrate features of similar aspects (e.g., price-related features), we organize the features by aspect (as shown in Table 4). Features within the same aspect are independently normalized using Min-Max normalization within each window (Schnaubelt, 2022). The normalization formula is as follows:

$$X_{norm,t,n} = \frac{X_{t,n} - X_{min,t,n}}{X_{max,t,n} - X_{min,t,n}} \tag{23}$$

where $X_{t,n}$ represents the feature of aspect n at time t. $X_{max,t,n}$ and $X_{min,t,n}$ denote the maximum and minimum values of the feature in aspect n within the sliding window up to time t. This approach ensures that normalization remains causal, preserving the integrity of the model's real-time predictions.

4.3. Feature selection

After addressing missing values, we utilized the basic features, limit order book features, and technical indicators of the 18 qualified stocks as inputs for predicting mid-price changes using the XGBoost model. The model achieved an average accuracy of 67.06% in predicting mid-price trends across all securities. Based on the feature importance derived from the model, we averaged and ranked the features, as illustrated in Fig. 4. The top 10 features in terms of importance were selected: bid-ask spread, mid-price changes, trading depth, trading volatility, Ask Depth, mid-price, best bid price, moving average line of the Bollinger Bands, upper bound, and lower bound. These features were used as the results of the feature selection process and as inputs for the price spread prediction module. In the following section, we will utilize all features and the selected features as inputs to predict price spread trends, further investigating whether applying feature selection techniques enhances the model's predictive performance.

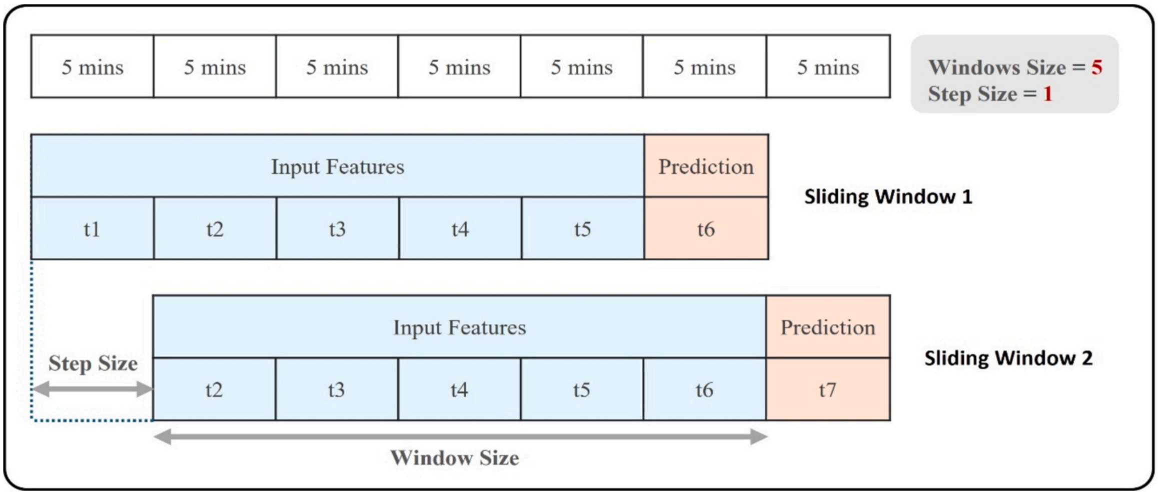

Fig. 3. Sliding window method.

Table 4
Summary of feature aspects.

Feature aspect	Features
Price	*Open, High, Low, Close, Mid Price, Best Bid Price, Best Ask Price, Middle Bollinger Band (MA), Upper Bollinger Band (Upper), Lower Bollinger Band (Lower), Avg Best Bid Price, Avg Best Ask Price, EMA*
Bid-ask spread	*BAS*
Depth	*Bid Depth, Ask Depth, Trading Depth*
Volume	*Volume, Bid Volume, Ask Volume*
Trading volatility	*TV*
Mid-price change	*Mid Price Change*
Price spread	*Price Spread*
RSI	*RSI*
KD	*K, D*
Distance of price spread and upper bound	$\mathrm{Dist}_t^{\mathrm{Upper}}$
Distance of price spread and lower bound	$\mathrm{Dist}_t^{\mathrm{Lower}}$
Distance of price spread and mean price spread	$\mathrm{Dist}_t^{\mathrm{Mean}}$

4.4. Pair selection

4.4.1. Correlation test

We conducted a correlation test on the mid-prices of the 18 previously selected stocks at 5-min intervals using the Pearson correlation coefficient. Stock pairs with a correlation coefficient greater than 0.9 were selected, resulting in 22 qualifying pairs, as shown in Table 4. The next step involves performing the Engle-Granger two-step cointegration test to determine whether a cointegration relationship exists between the selected stock pairs.

4.4.2. Cointegration test

According to the results in Table 5, we first conducted the ADF test on the mid-price series of each stock. The results indicated that the mid-price series of all stocks were above the significance level, suggesting that all mid-price series are non-stationary. Therefore, we further performed the Engle-Granger two-step cointegration test on the mid-price series of stock pairs with a correlation coefficient greater than 0.9 to determine whether there is a cointegration relationship between the non-stationary mid-price series of the two stocks in each pair. We used a significance level of 0.1 for the Engle-Granger two-step test. Although 0.05 is a more common significance level, we believe a more lenient level can help identify more potential stock pairs when searching for and testing long-term equilibrium relationships between stocks, increasing the likelihood of detecting true cointegration relationships. The test results in Table 6 indicate that there are 6 stock pairs with a cointegration relationship, which fits the assumption of pairs trading. Therefore, we selected EMC (2603) and WEC (2344), EMC (2603) and PANJIT (2481), NTC (2408) and WEC (2344), PANJIT (2481) and TSMC (2330), NTC (2408) and Macronix (2337), and CAL (2610) and EMC (2603) as the pairs for pairs trading.

4.5. Evaluation metrics

4.5.1. Price spread prediction

Evaluating the model's performance solely on accuracy is insufficient due to the highly imbalanced class distribution. To better assess the model's predictive ability across different categories (Category 1: price spread within the boundary, Category 2: abnormal price spread, and Category 3: price spread reverting to the mean), we employ Precision, Recall, and F1 Score as evaluation metrics. These metrics offer a more comprehensive evaluation of the model's effectiveness in handling each category.

4.5.2. Backtesting

In pairs trading backtesting, we evaluate the performance of the trading strategy using several key indicators: trading capital, cumulative profit, return rate, average profit and loss, number of pairs traded, win rate, Maximum Drawdown (MDD), Sharpe Ratio, and Sortino Ratio. Trading capital refers to the total investment required throughout the trading period, while cumulative profit represents the total profit accumulated over this time. The number of pairs traded indicates how frequently a pair of stocks were bought and sold, completing an entire trading cycle, as illustrated in Fig. 5. Average profit is calculated by dividing the total profit by the number of pairs traded, representing the average profit per trade. The win rate is calculated by dividing the number of profitable trades by the total number of trades. MDD measures the most significant peak-to-trough decline in trading capital, expressed as a percentage.

The Sharpe Ratio measures the return per unit of risk and is calculated as follows:

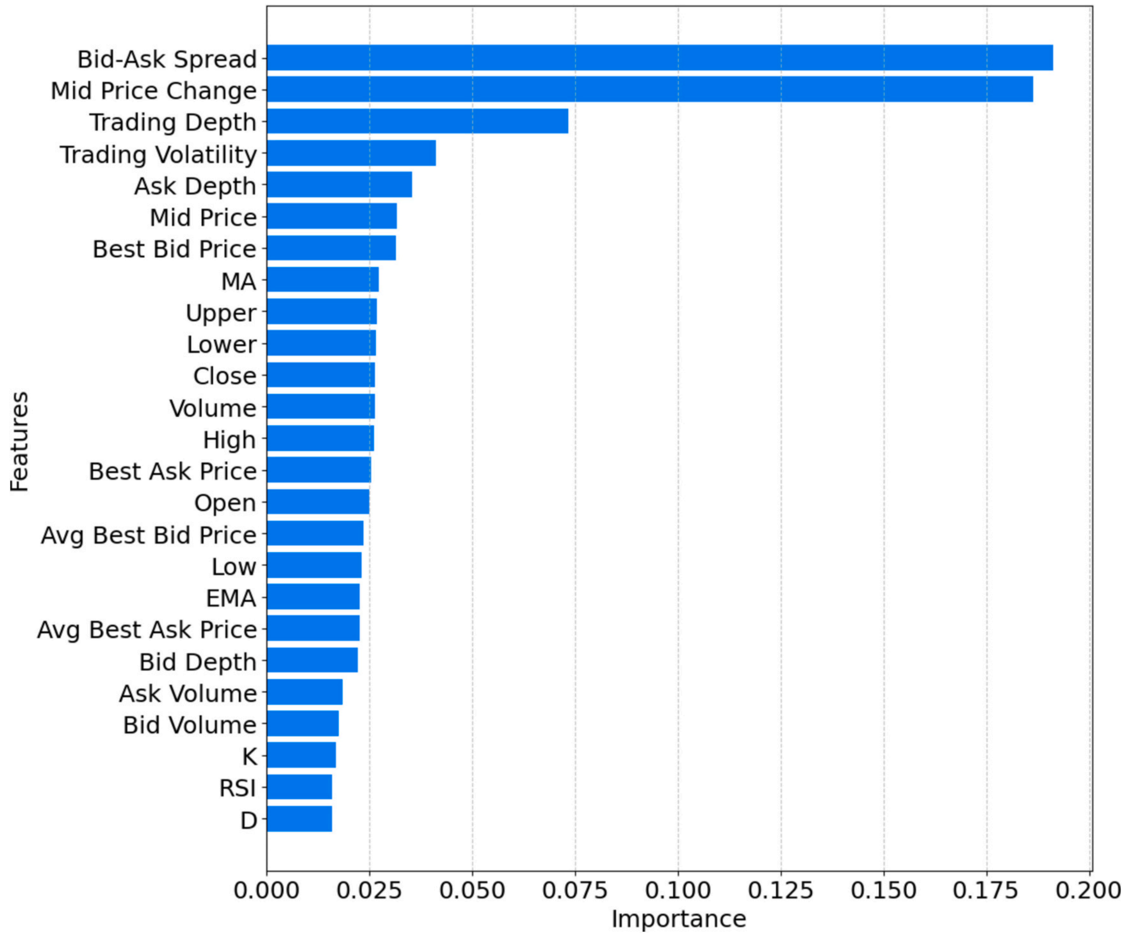

Fig. 4. Feature selection results.

Table 5
Stock pairs with a correlation coefficient greater than 0.9.

Stock 1	Name	Stock 2	Name	Correlation
2609	YMTC	2603	EMC	0.983870
2615	WANHAI	2603	EMC	0.970521
2603	EMC	2344	WEC	0.948651
2330	TSMC	2303	UMC	0.948517
2615	WANHAI	2303	UMC	0.947889
2615	WANHAI	2344	WEC	0.944732
2603	EMC	2481	PANJIT	0.943622
2603	EMC	2303	UMC	0.942476
2615	WANHAI	2609	YMTC	0.941018
2408	NTC	2344	WEC	0.940873
2481	PANJIT	2330	TSMC	0.940742
2481	PANJIT	2303	UMC	0.939548
2344	WEC	2303	UMC	0.937491
2609	YMTC	2344	WEC	0.934970
2615	WANHAI	2481	PANJIT	0.925934
2401	SUNPLUS	2330	TSMC	0.924334
2408	NTC	2337	Macronix	0.923315
2401	SUNPLUS	2303	UMC	0.921900
2609	YMTC	2481	PANJIT	0.920311
2609	YMTC	2605	SNC	0.909833
2609	YMTC	2303	UMC	0.904049
2610	CAL	2603	EMC	0.900451

Table 6
Results of Engle-Granger two-step cointegration test.

Stock 1	Stock 2	*P*-value	Cointegration relationship
2609	2603	0.1962	None
2615	2603	0.1666	None
2603	2344	0.0854	**Yes**
2330	2303	0.2741	None
2615	2303	0.2225	None
2615	2344	0.1825	None
2603	2481	0.0703	**Yes**
2603	2303	0.2644	None
2615	2609	0.6559	None
2408	2344	0.0031	**Yes**
2481	2330	0.0949	**Yes**
2481	2303	0.1173	None
2344	2303	0.6527	None
2609	2344	0.1417	None
2615	2481	0.2035	None
2401	2330	0.5236	None
2408	2337	0.0001	**Yes**
2401	2303	0.6506	None
2609	2481	0.4376	None
2609	2605	0.3475	None
2609	2303	0.6680	None
2610	2603	0.0088	**Yes**

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \tag{24}$$

where R_p is the average return of the pairs trading, R_f is the risk-free rate (assumed to be 0 in this study), and σ_p is the standard deviation of the pairs trading returns. The Sortino Ratio, similar to the Sharpe Ratio, measures return per unit of risk but focuses solely on downside risk, emphasizing the potential losses associated with the investment strategy.

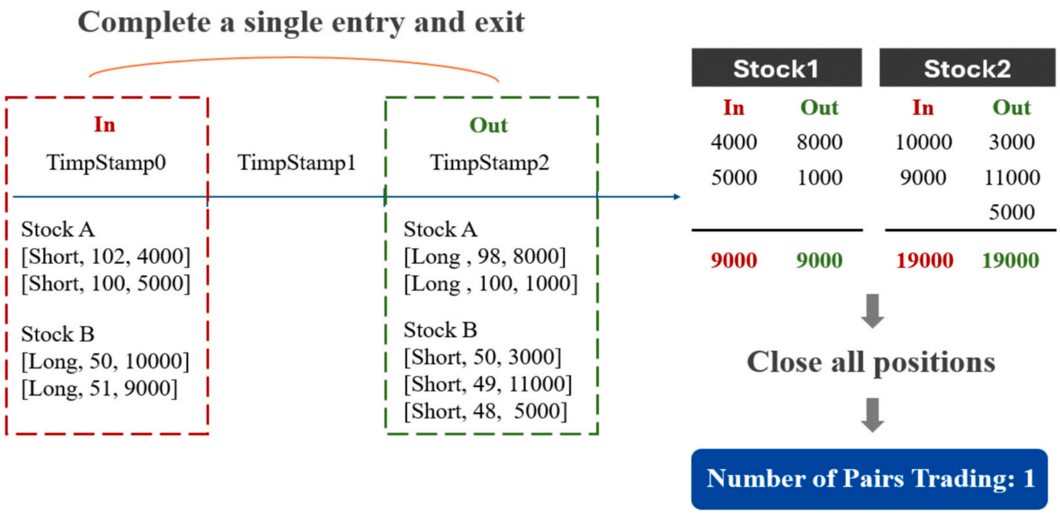

Fig. 5. Diagram of pairs trading frequency calculation.

5. Results and evaluation

5.1. Price spread prediction

Building on the feature selection results, we applied pairs trading to the stock pairs exhibiting a cointegration relationship. The model was employed to predict the entry and exit points for pairs trading, assessing its ability to generate profits while mitigating systemic risks.

Table 7 presents the results of the price spread prediction module during the backtesting period. The best-performing model varies across stock pairs, indicating that no single deep-learning model consistently delivers optimal prediction performance for all pairs. Instead, the model selection should be based on the specific characteristics of each stock pair to achieve the best predictive results. Evaluation indicators across different categories reveal that, regardless of the model, the prediction performance for Category 2 (abnormal price spread) remains consistently poor.

To assess whether the model struggles to capture the price spread trend, we analyzed the determination prediction rates of the best-performing models for each stock pair across categories. Determination prediction refers to instances where the model predicts the probability of the mid-price trend at the next time point falling into a specific category with a confidence level greater than 0.7 based on the features at that time. Suppose the model has not effectively converged during training. In that case, the probabilities assigned to each category in the test set will be nearly identical, leading to predictions driven by slight differences in probability. For example, if the model predicts the probabilities for each category as [0.33, 0.35, 0.32], it will predict Category 2.

Therefore, we define the determination prediction rate as a measure of model convergence, calculated as follows:

$$\text{Determination Prediction Rate} = \left(\frac{\text{Number of predictions where the probability for Category n} > 0.7}{\text{Total number of predictions for Category n}}\right) \times 100\% \tag{25}$$

From Table 8, we observe that the model's determination prediction rate for each category remains above 80%, indicating that the model successfully learns the pattern of price spread changes from the features of the stock pairs. We then examined the entry and exit timing predicted

Table 7
Results of price spread prediction.

Stock pair	Best model	Evaluation metrics	Category 1	Category 2	Category 3
2603 2344	LSTM-Attention	Precision	0.9487	0.2308	0.7942
		Recall	0.9545	0.2927	0.7154
		F1 Score	0.9516	0.2581	0.7527
2603 2481	BiLSTM	Precision	0.9665	0.5046	0.3750
		Recall	0.9585	0.5392	0.4615
		F1 Score	0.9625	0.5213	0.4138
2408 2344	BiGRU	Precision	0.9620	0.3934	0.7513
		Recall	0.9322	0.3721	0.9367
		F1 Score	0.9469	0.3825	0.8338
2481 2330	BiLSTM	Precision	0.9291	0.3152	0.3704
		Recall	0.9003	0.3688	0.6667
		F1 Score	0.9145	0.3399	0.4762
2408 2337	LSTM-Attention	Precision	0.9725	0.3043	0.6667
		Recall	0.8934	0.3984	0.9219
		F1 Score	0.9313	0.3451	0.7738
2610 2603	BiGRU	Precision	0.9568	0.2360	0.7924
		Recall	0.9172	0.2734	0.8151
		F1 Score	0.9366	0.2533	0.8036

Table 8
Results of model deterministic prediction rate.

Stock pair	Category 1	Category 2	Category 3
2603 2344	98.78%	86.54%	93.33%
2603 2481	99.25%	95.41%	100%
2408 2344	99.70%	92.62%	97.46%
2481 2330	97.70%	83.64%	96.30%
2408 2337	99.59%	96.89%	98.87%
2610 2603	99.08%	86.96%	98.73%

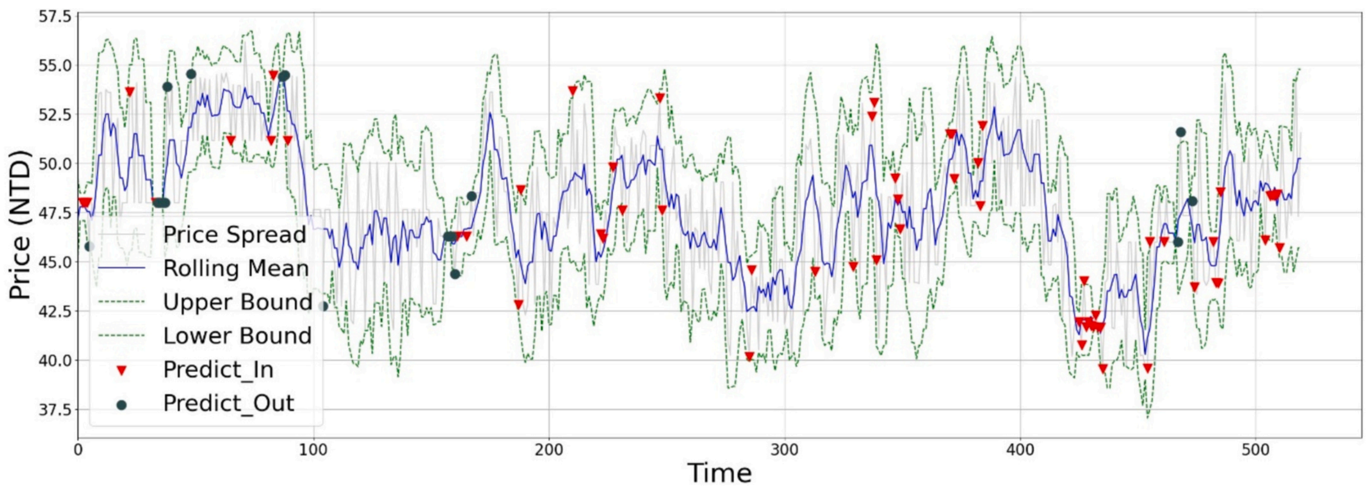

Fig. 6. Model's predicted entry and exit timing (example: stock pair 2408 and 2337).

Table 9
Results of price spread prediction (without feature selection).

Stock pair	Evaluation metrics	Category 1	Category 2	Category 3
2603 2344	Precision	0.9332	0.1841	0.8246
	Recall	0.9523	0.3274	0.5957
	F1 Score	0.9427	0.2357	0.6917
2603 2481	Precision	0.9676	0.3643	0.3333
	Recall	0.9216	0.5426	0.5385
	F1 Score	0.9440	0.4359	0.4118
2408 2344	Precision	0.9677	0.2328	0.6486
	Recall	0.8200	0.4615	0.7742
	F1 Score	0.8877	0.3095	0.7059
2481 2330	Precision	0.9145	0.2278	0.2941
	Recall	0.8774	0.2975	0.3571
	F1 Score	0.8956	0.2581	0.3226
2408 2337	Precision	0.9650	0.2599	0.6709
	Recall	0.8733	0.4000	0.8548
	F1 Score	0.9168	0.3151	0.7518
2610 2603	Precision	0.8864	0.2026	0.7620
	Recall	0.9004	0.2403	0.6894
	F1 Score	0.8933	0.2199	0.7239

by each model. For instance, with the stock pair NTC (2408) and Macronix (2337) (as shown in Fig. 6), the red triangles represent the model's predicted entry points, while the green dots represent the predicted exit points. Although the actual price spread falls within the boundary at specific time points, the model predicts it as an entry point (suggesting an abnormal price spread). Further analysis shows that while the model's predicted entry and exit timing is generally accurate, prediction errors may stem from the rapid and substantial fluctuations in the mid-price of stocks every five minutes. These fluctuations make it challenging for the model to fully grasp the relationship between the price spread and the upper and lower boundaries. Even though the model predicts an entry point, the actual price spread within the boundary may be approaching the boundary. Hence, while the deep learning model may not always precisely predict the relationship between the price spread and the boundaries, it can still capture the overall trend of price spread changes.

We also conducted price spread prediction without feature selection, utilizing all available features to assess the necessity of feature selection. The results, presented in Table 9, were compared with those using selected features (Table 7). The comparison reveals that models employing feature selection outperformed those that did not, confirming that using XGBoost for feature selection effectively filters out noisy stock market features. This, in combination with deep learning models, enhances predictive performance.

Table 10
Backtest results.

Backtesting period	2020/02/24–2021/03/31
Stock pairs	6 pairs
Trading capital (NT dollars)	6,572,633
Cumulative profit (NT dollars)	725,483
Return rate (%)	11.0379
Average profit (NT dollars)	7328
Number of pairs trading	99
Win rate (%)	68.6869
Sharpe ratio	0.8784
Sortino ratio	5.4606
MDD (%)	0.4825

5.2. Backtesting

We used the model developed in the previous section to predict price spreads for each stock pair, so we conducted a backtest of the pairs' trading strategy. The strategy involves shorting the stock with a higher mid-price and buying the stock with a lower mid-price at the time of entry. The backtest was structured such that each position must be closed before initiating a new trade for the same pair. Following Taiwan's stock market regulations, the trading cost was set at 0.3%. The experiment assumes no market competition (i.e., trades are matched without influence from other traders) and that all trades do not impact market price movements.

As shown in Table 10, the backtest results reveal that using the model's predicted entry and exit points for pairs trading in stock pairs with a cointegration relationship results in a win rate of 68.69% across 99 trades involving six cointegrated stock pairs. Additionally, the strategy performed well on various risk-related indicators, with a notable Maximum Drawdown (MDD) of 0.4825%, demonstrating its effectiveness in mitigating substantial loss risks. The results confirm that the proposed framework maintains a consistent win rate and return, even during the COVID-19 pandemic, effectively reducing investor loss risks. This suggests combining deep learning models with pairs trading can manage risk and achieve high profitability in volatile intraday markets.

5.3. Research comparisons

We conducted a detailed comparative analysis of studies utilizing LSTM and Transformer architectures (Barez et al., 2023; Bilokon and Qiu, 2023; Kolm et al., 2023). Table 11 provides a summary of these comparisons, focusing on high-frequency trading applications, market

Table 11
Research comparisons.

Aspect	Our study	Barez et al. (2023)	Bilokon and Qiu (2023)	Kolm et al. (2023)
High-frequency	Yes	Yes	Yes	Yes
Market	Taiwan stock	Cryptocurrency	Cryptocurrency	Nasdaq stock
Features	Basic features, limit order book, technical indicators	Limit order book	Basic features, limit order book, technical indicators	Limit order book
Pair trading	Yes	Yes	Not specified	Not specified
Models used	LSTMs	LSTMs vs. Transformers	LSTMs vs. Transformers	LSTMs
Results	Best LSTM model varies across stock pairs	LSTMs outperform transformers for LOB data	LSTMs outperform transformers	LSTMs outperform other architectures

focus, feature sets, pair trading considerations, models used, and results. Key points of comparison include:

(1) Market: The studies cover different markets. Our study focuses on Taiwan stocks, which exhibit unique characteristics, such as lower volatility and a more regulated environment, in contrast to the cryptocurrency and Nasdaq markets, which are highly volatile and less regulated.
(2) Features: Our study, along with that of Bilokon and Qiu (2023), incorporates a more comprehensive set of basic features, Limit Order Book (LOB) features, and technical indicators than other studies, which primarily emphasize specific aspects of LOB data. Our feature set is broader than that used by Bilokon and Qiu (2023). Additionally, we employ the XGBoost model for feature selection, which was not applied in their study.
(3) Pair trading: Pair trading strategies are examined in our study and in the work of Barez et al. (2023), while this aspect is not specified in the other studies. In our pair selection module, we utilize the Pearson correlation coefficient, the Augmented Dickey-Fuller test, and the Engle-Granger two-step approach for pair selection, which Barez et al. (2023) does not implement.
(4) Model comparison: Across all studies, LSTM models consistently outperform Transformers and other architectures, particularly in handling noise-prone LOB data. In our study, we found that selecting the best model depends on the specific characteristics of each stock pair, a consideration not emphasized in other studies.

6. Conclusion

6.1. Discussion

Traditionally, stock market predictions have focused on forecasting daily closing prices. However, given the high frequency and volatility of stock market transactions, daily intervals may overlook significant intraday information. Furthermore, there is limited research on applying deep learning models to predict the relationship between price spreads and boundaries for pairs trading signals. This study, from the perspective of institutional investors, utilizes intraday continuous trading data from the Taiwan Stock Exchange, covering the period from April 1, 2020, to March 31, 2021, to examine whether combining pairs trading with deep learning models can effectively mitigate risk and generate profit in the Taiwan stock market during the COVID-19 pandemic. The key contributions of this study are as follows:

1. We utilized intraday continuous trading data to integrate three stock market features: basic features, limit order book features and technical indicators. Using the XGBoost model to select the top 10 essential features significantly contributing to predicting the mid-price, we found that predictions with feature selection outperformed those without.
2. We identified stock pairs with high correlation and cointegration relationships through correlation and cointegration tests. The essential features of these stock pairs were then used as inputs for the deep learning model. By predicting the relationship between the price spread and boundaries, we provided appropriate entry and exit signals for pairs trading. Although accurately predicting the relationship between price spreads and boundaries remains challenging, the deep learning model demonstrated an ability to learn the patterns of price spread changes.
3. We applied the model's predicted entry and exit signals to pairs trading backtesting. The results indicated that even amidst the impacts of the COVID-19 pandemic and accounting for trading costs, the pairs trading strategy could maintain a high win rate and profit while mitigating risks. This confirms that our framework can achieve stable profits in a high-frequency, volatile intraday trading environment.

6.2. Limitation and future work

This study focused solely on stock pairs with high positive correlation for selection, resulting in a limited number of pairs exhibiting high correlation and cointegration. In practice, investors also utilize pairs with high negative correlation, simultaneously buying and selling them for pairs trading. However, we did not consider negatively correlated pairs, which restricted our ability to demonstrate the effectiveness of trading across a broader range of stock pairs. Additionally, due to restrictions on accessing recent data from the Taiwan Stock Exchange, our study covered only the pandemic period, with a relatively short trading duration, and we did not test the framework across different periods. Moreover, our research was limited to data from the Taiwanese stock market, while pairs trading can also involve assets from various financial markets.

Future research could incorporate data from a broader range of periods, capturing different market conditions such as bear markets, bull markets, and periods of high volatility. This would enable the model to adapt to diverse conditions and provide more accurate entry and exit signals. Furthermore, expanding the dataset to include assets from different financial markets, such as futures and options, could diversify the investment portfolio and profit strategies, potentially enhancing profit opportunities and returns. Moreover, in highly volatile markets, such as cryptocurrency and major Western stock markets, future studies might also consider incorporating additional features, such as trade volume imbalance, best-order imbalance, and queue imbalance (Schnaubelt, 2022), to enrich model analysis. Furthermore, implementing the walk-forward technique with periodic recalibration during backtesting could improve forecasting accuracy and yield more reliable results by allowing the model to adapt to shifting market dynamics over time.

Data availability

No

Acknowledgments

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